Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 34

Permutations

Covers Chartrand & Zhang, Section 8.4

Status of this set

Ungraded practice. Solutions released with it. Show the expression, not just the number.

Part I: How to Practice This Material

The formula and where it comes from

$$P(n,r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

This is the Multiplication Principle with one fewer choice at each stage. If you forget the formula, rebuild it: n choices, then $n-1$, and so on for r stages.

The test

Does reordering the chosen items change the outcome? If yes, it is a permutation.

Three mistakes to expect

1. **Using $P(n,r)$ when order does not matter.** That over-counts by $r!$.
2. **Ignoring a restriction** — a leading digit, a fixed position.
3. **Giving a bare number.** The expression is the reasoning.

Part II: Practice Problems

Section A — Direct counts

1. Compute each, showing the product.

(a) $P(9,3)$ (b) $P(6,6)$ (c) $P(12,2)$

2. In how many ways can 7 distinct books be arranged on a shelf? In how many ways if two specific books must be adjacent?

Section B — Restrictions

3. How many four-letter strings can be formed from the letters of NUMBERS with no repeated letter? How many begin with a vowel?

4. A club of 10 members elects a president, a vice-president, and a treasurer, with no person holding two offices.

(a) How many outcomes are possible?

(b) How many if a particular member must be president?

(c) How many if a particular member must hold *some* office?

Section C — Writing

5. In four to six complete sentences, explain why $P(n, r) = \dfrac{n!}{(n-r)!}$ follows from the Multiplication Principle. Say what $(n-r)!$ represents.

Section D — Challenge (optional)

6. In how many ways can 5 people be seated around a *circular* table, if two seatings are the same when one is a rotation of the other?